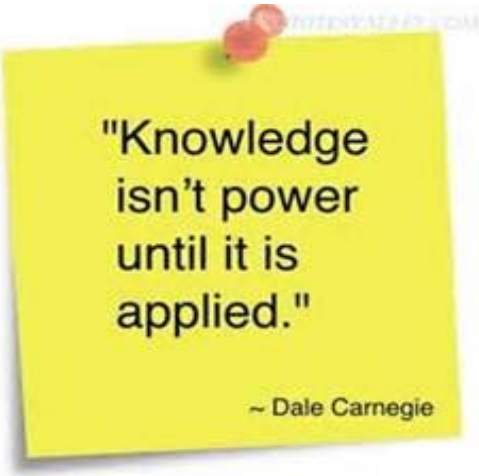
Fall 2019 VE496
Lecture notes
Week 1

Tips for Succeeding in this Course:

- Show up
- Keep up
- Review assigned materials
- Apply what you learn



All images taken from http://cn.bing.com/images/search

Technical Communication basics:

✓ Determine your purpose.

✓ Assess your audience's needs.

✓ Consider the context of your writing.

- **Establish** the scope of your coverage on a topic
- **Gather** facts
- **Organize** your ideas & thoughts
- **Select** the appropriate medium

Types of Technical Communication:

Oral: Job interviews, speeches, presentations, meetings

Written: Manuals, proposals, reports, research papers, resumes/CV, job applications, etc.

Electronic: Emails, text messages, videoconferencing, podcasts, blogs

Visual: Symbols, icons, tables, figures, drawings, illustration, photographs, video, etc.

Profiling your audience:

- Who is your audience?
- What are your audience's needs in relation to the subject?
- Have you narrowed your topic to best focus on what you want your audience to know?

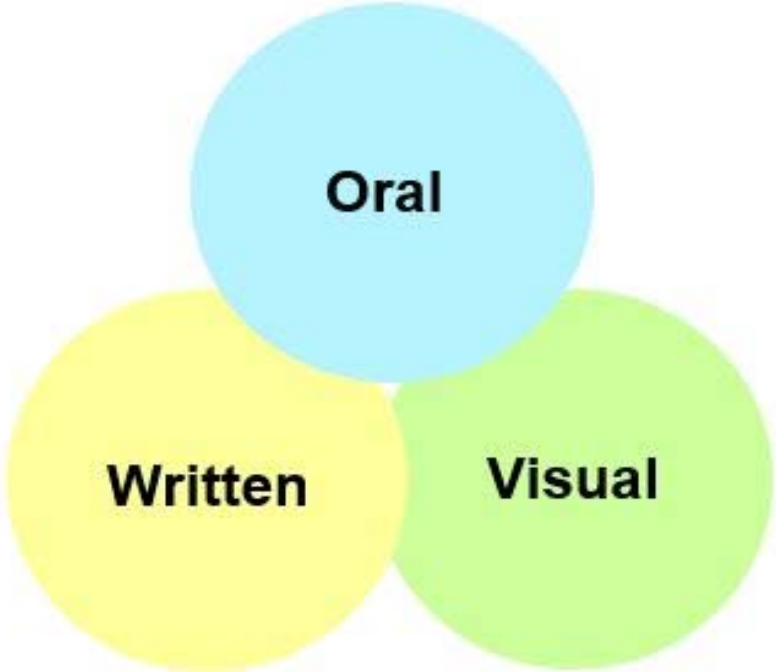
Technical Communication → **- Organizing information**

- Motivating action

- Interactive & adaptable
- Reader centered
- Reliant on teamwork
- Visual
- Bound ethically & legally
- International & cross-cultural

Multi-modality

Technical communication is a design process using words, images, and voice.



Topics of interests:

- Data visualization
- Statistical graphics
- Information design

✓ **Understand-*ability***

✓ **Use-*ability***

“Can't see the forest for the trees”

Looking at things one at a time, you might not realize that a branch of separate "trees" go together to make a "forest."

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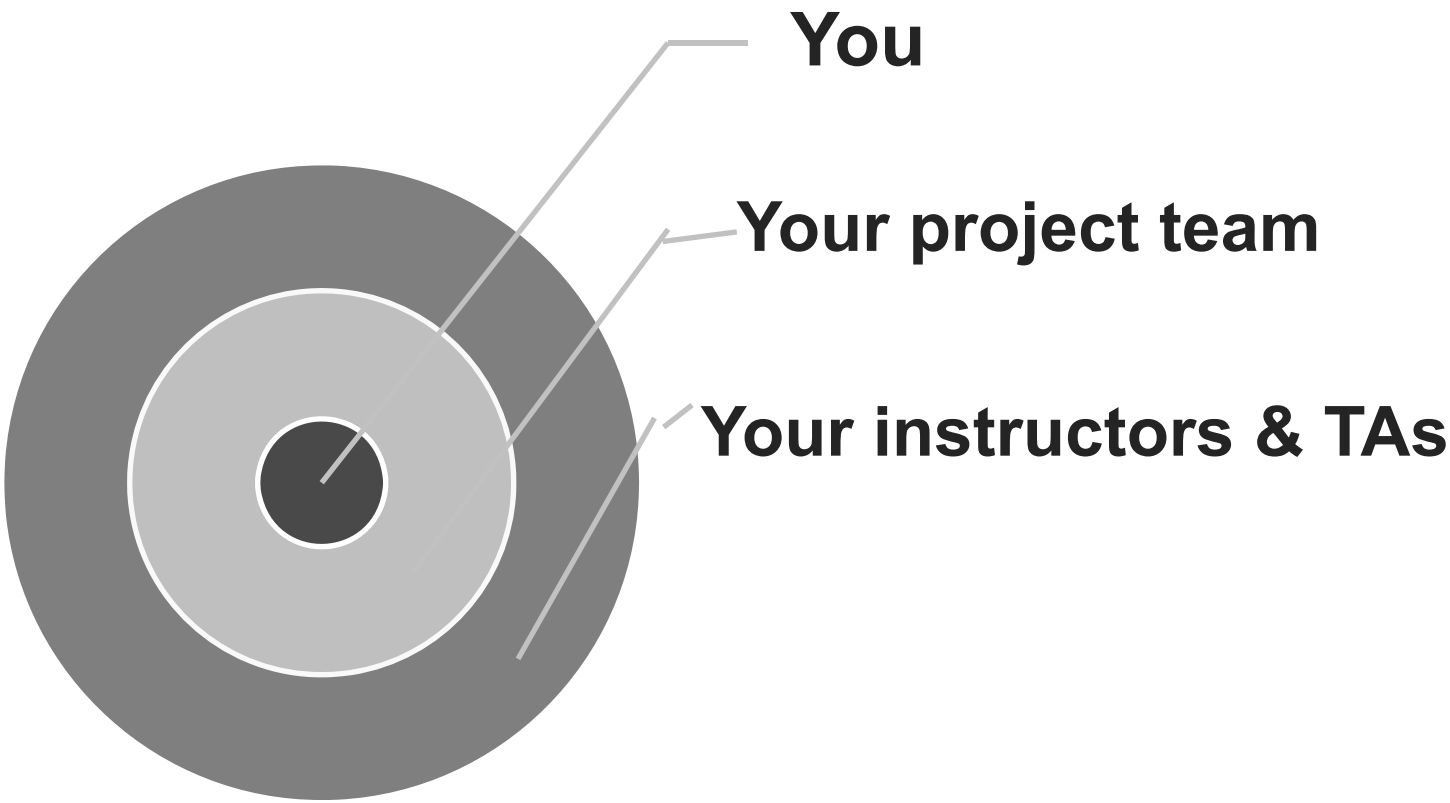
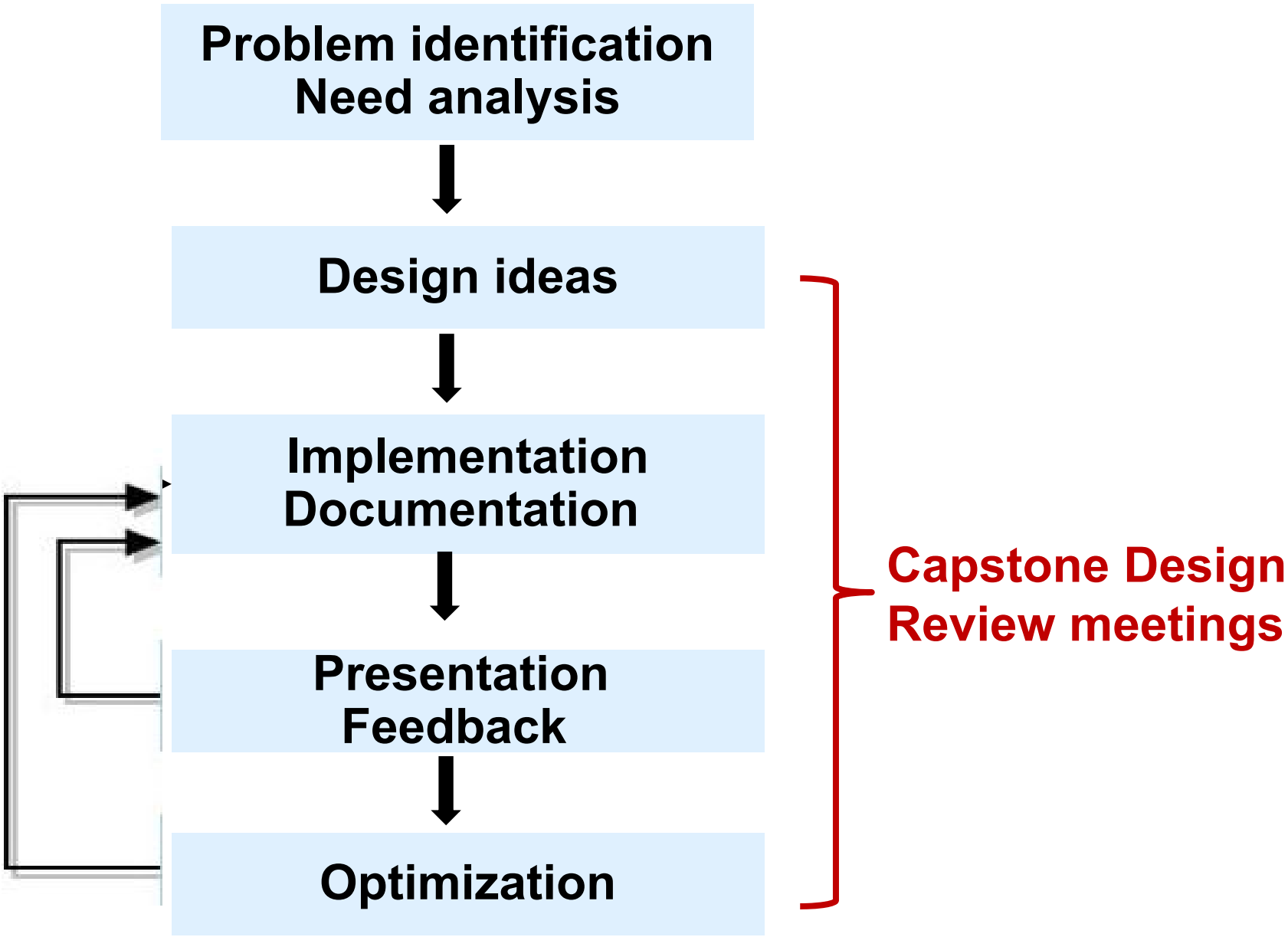
All images taken from http://cn.bing.com/images/search

When life is easy, your audience looks like this:

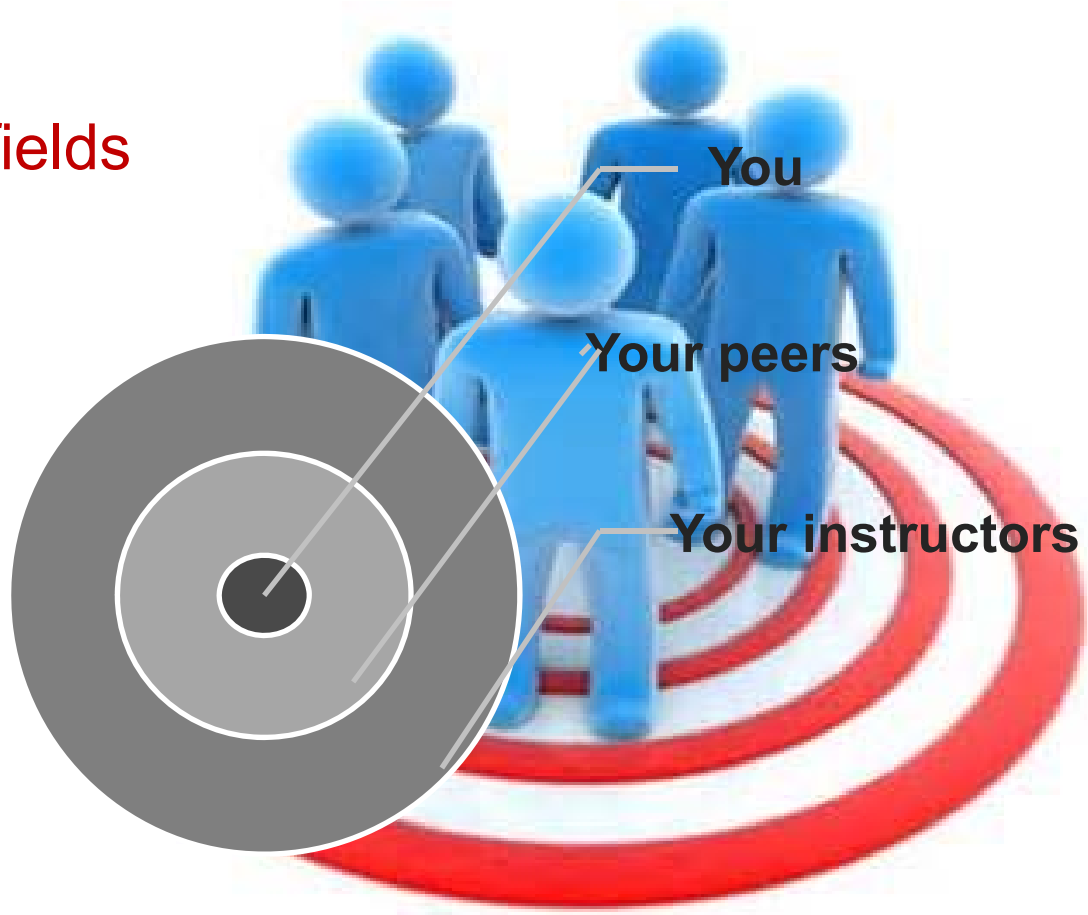



Problem identification
Need analysis
Design ideas
Implementation
Documentation
Presentation
Feedback
Optimization
Capstone Design
Review meetings

For tech com → Wider audiences:

- Potential clients
- Marketing
- Sales
- Engineers in different fields
- Government agencies
- Customers

- People **outside**
  Science
  Technology,
  Engineering
  Math (**STEM**)



Innovation & commercialization

Input

- Social concerns
- Customers
- Needs/demands
- Materials
- Capital
- Time/energy
- Talent
  - knowledge
  - skills

Process

- Designing
- Planning
- Producing & constructing
- Managing
- Marketing
- Financing
- Documenting

Output

- Products, systems structures for various markets
- Support activities
  - Training
  - Service
  - Customer satisfaction
- Company profits

Today's engineering students entering industry will spend more time explaining technology to different stakeholders.

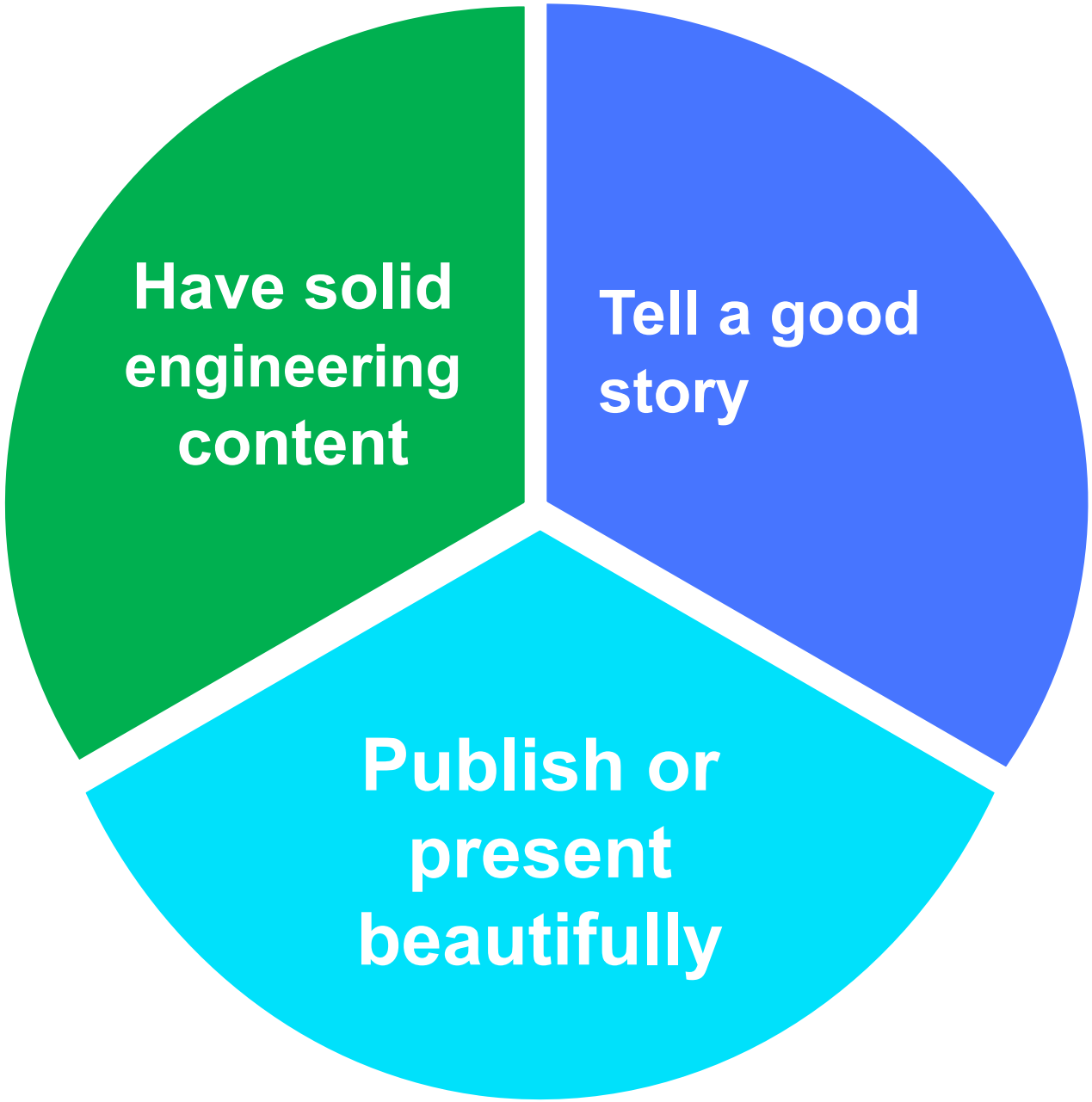
- Consumers
- Legislators
- lawyers
- Judges
- Bureaucrats
- Environmentalists
- The media

Why communicate?

- To build *common grounds of understanding* with audience

What is your purpose or what do you want to achieve with your audience?

- To help them learn about a subject or problem
- To persuade them to see a problem through your eyes and accept your solution for it
- motivate them to take action


Have solid engineering content
Tell a good story
Publish or present beautifully

Post graduation plans

- Job search
- Graduate school
- Internship
- Gap year

Preparing Graduation school application

- Short-list your graduate programs
- Make a timeline
- Create a check-list of required documents
- Follow instructions, requirements, word count, etc.
- Ask for letters of recommendation

International students in the US: What is the “big picture” ?

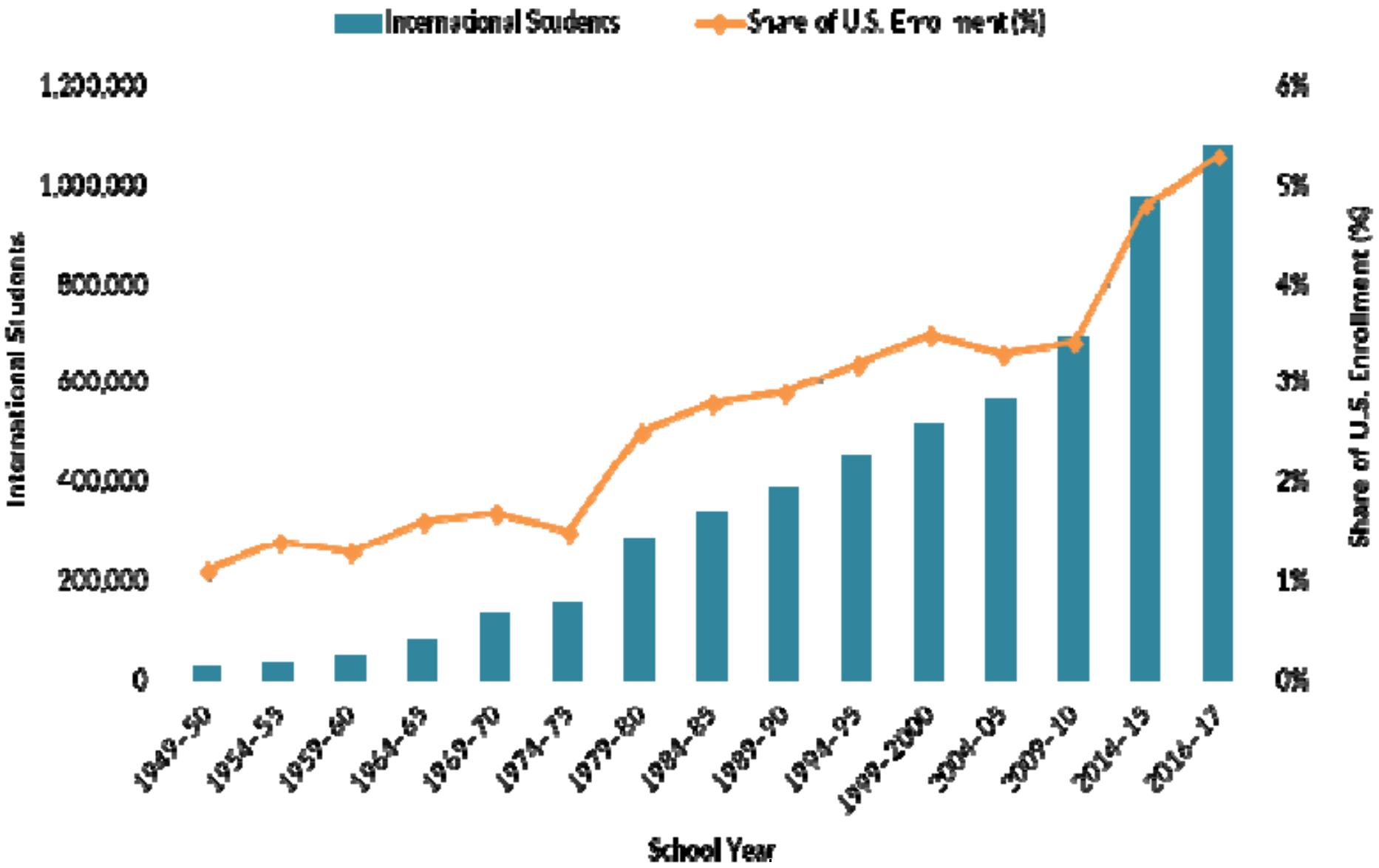
Migration Policy Institute

https://www.migrationpolicy.org/article/international-students-united-states#EnrollmentTrends

The Institute of International Education (IIE)

https://www.iie.org/

International Students in U.S. Colleges and Universities and Share of Total Enrollment, (%), School Year (SY) 1949–50 to 2016–17



Newly enrolling students decreased in starting Fall 2016

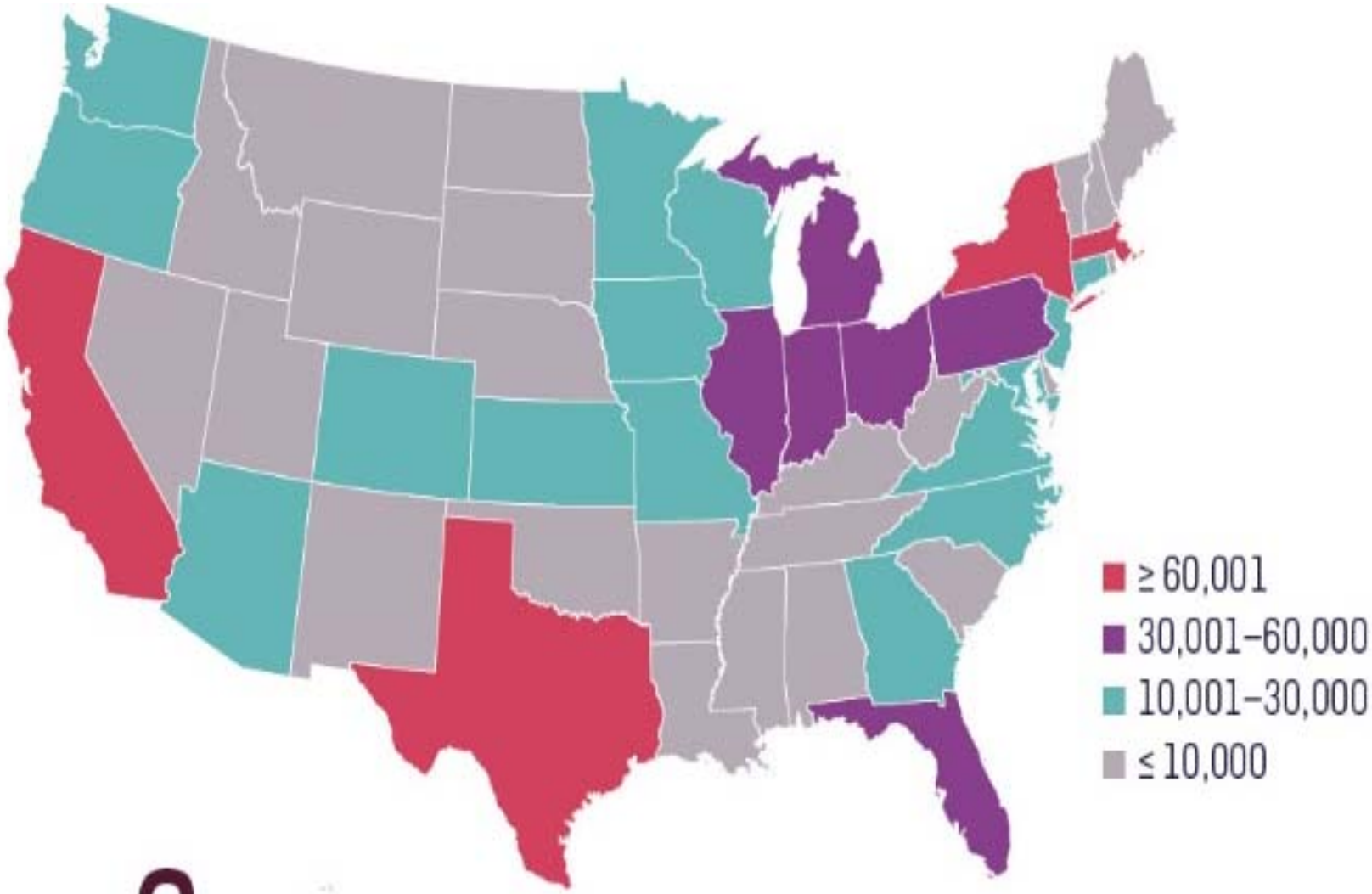
Downward trend of new international student enrollment continues, but at a less steep rate.

- **Visa delay & denial**
- **Costs of U.S. higher education,**
- **Shifting social and political climate**
- **Competition from institutions in other countries**
- **Concerns about job prospects in US**

Top Ten States by International Student Population, SY 2016–17

State	International Students		Total Immigrant Population	
	Number	Share of U.S. Total (%)	Number	Share of U.S. Total (%)
United States	**1,079,000**	**100.0**	**43,739,300**	**100.0**
California	157,000	14.5	10,677,700	24.4
New York	118,000	11.0	4,536,100	10.4
Texas	85,000	7.9	4,729,900	10.8
Massachusetts	63,000	5.8	1,123,900	2.6
Illinois	52,000	4.8	1,783,500	4.1
Pennsylvania	51,000	4.7	870,900	2.0
Florida	46,000	4.2	4,236,500	9.7
Ohio	39,000	3.6	513,600	1.2
Michigan	34,000	3.2	662,300	1.5
Indiana	31,000	2.8	349,200	0.8
Other States	403,000	37.3	14,255,700	32.6

US destinations of international students



1 out of 3 international students

studies in California, New York, or Texas

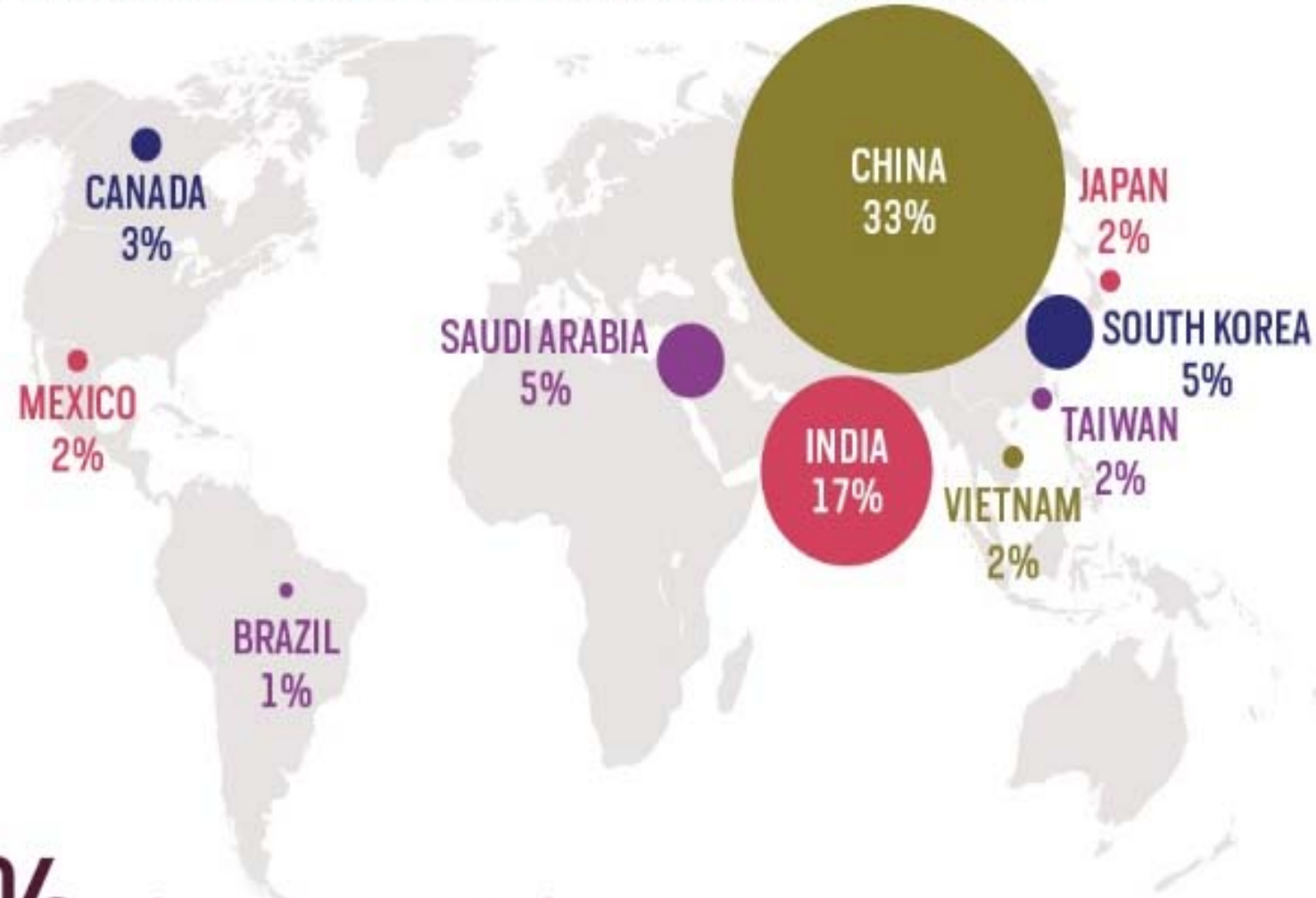
TOP U.S. INSTITUTIONS HOSTING INTERNATIONAL STUDENTS, 2016/17

1	New York University	New York	NY	17,326
2	University of Southern California	Los Angeles	CA	14,327
3	Columbia University	New York	NY	14,096
4	Northeastern University - Boston	Boston	MA	13,201
5	Arizona State University - Tempe	Tempe	AZ	13,164
6	University of Illinois - Urbana-Champaign	Champaign	IL	12,454
7	University of California - Los Angeles	Los Angeles	CA	12,199
8	Purdue University - West Lafayette	West Lafayette	IN	11,288
9	University of Texas - Dallas	Richardson	TX	9,305
10	Penn State University - University Park	University Park	PA	9,134
11	University of California - San Diego	La Jolla	CA	9,065
12	Boston University	Boston	MA	8,992
13	University of Michigan - Ann Arbor	Ann Arbor	MI	8,163
14	University of Washington	Seattle	WA	8,019
15	University of California - Berkeley	Berkeley	CA	8,000
16	Michigan State University	East Lansing	MI	7,779
17	Ohio State University - Columbus	Columbus	OH	7,684
18	Carnegie Mellon University	Pittsburgh	PA	7,653
19	Indiana University - Bloomington	Bloomington	IN	7,502
20	University of Texas - Arlington	Arlington	TX	7,277

of international students studying in the US in 2016/17, by country of origin

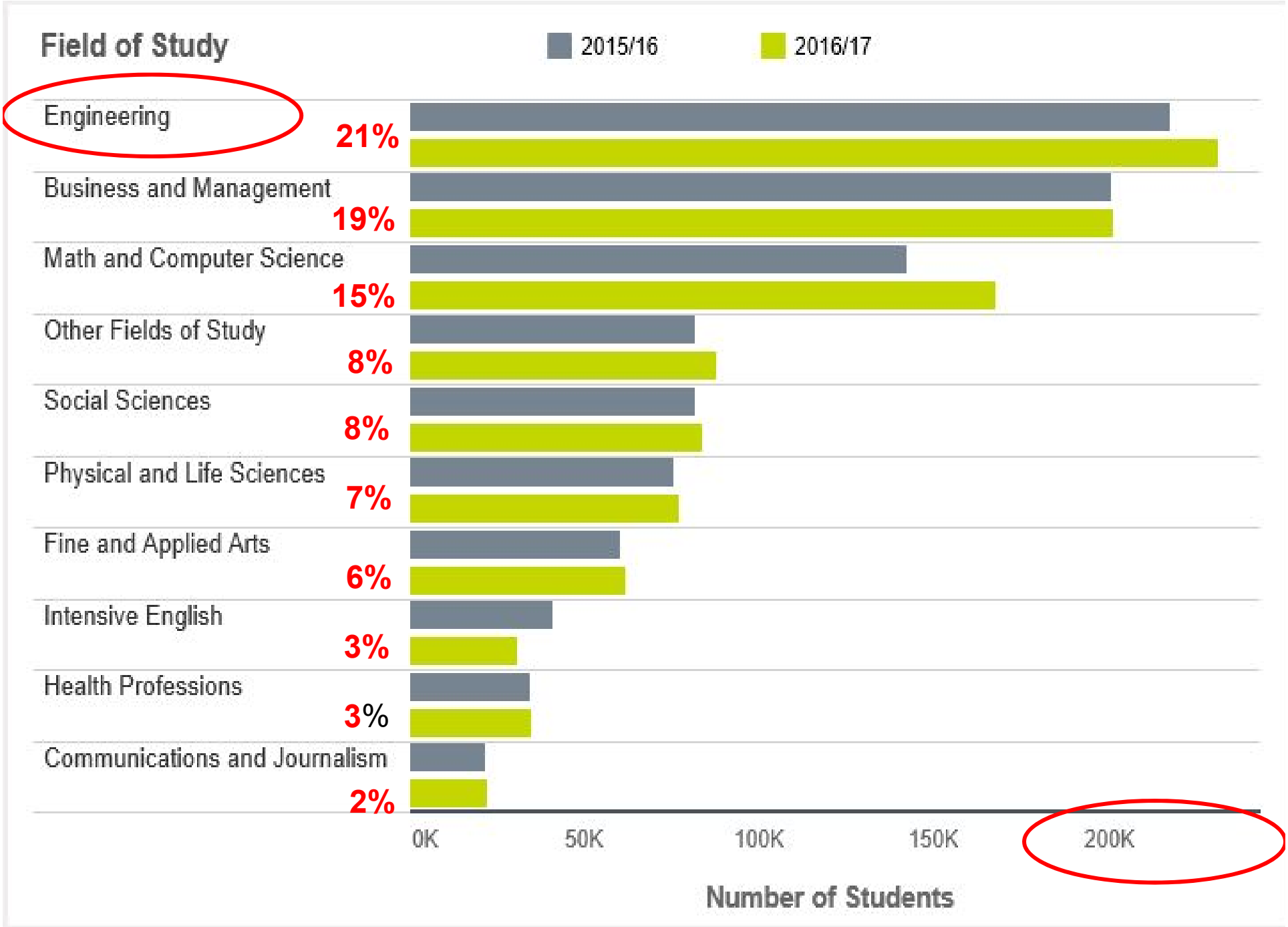
2016-17		
Country	Number	Share (%)
Total	1,079,000	100.0
China	351,000	32.5
India	186,000	17.3
South Korea	59,000	5.4
Saudi Arabia	53,000	4.9
Canada	27,000	2.5
Vietnam	22,000	2.1
Taiwan	22,000	2.0
Japan	19,000	1.7
Mexico	17,000	1.6
Brazil	13,000	1.2
Other Countries	311,000	28.8

TOP TEN PLACES OF ORIGIN OF INTERNATIONAL STUDENTS

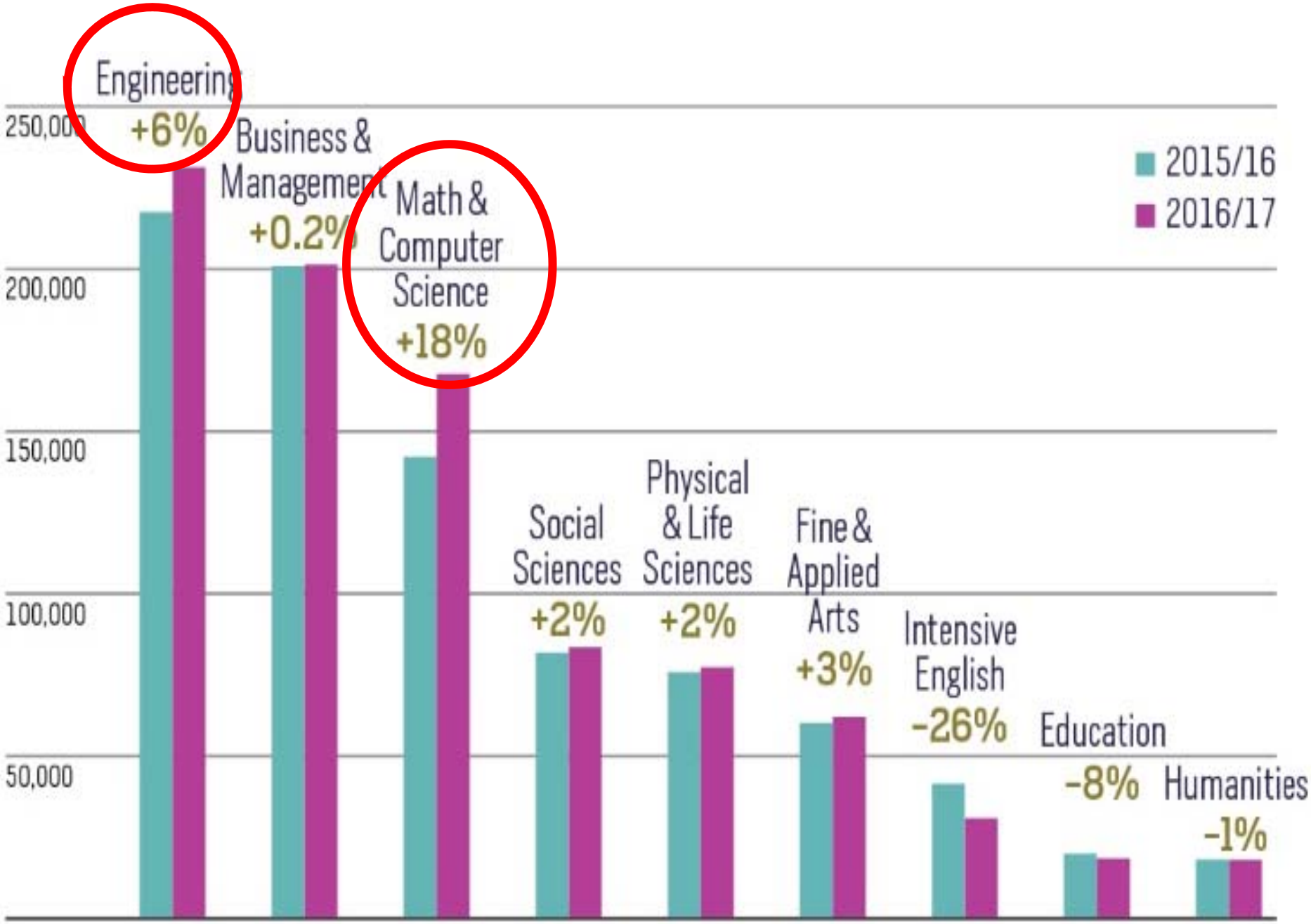


50% of international students

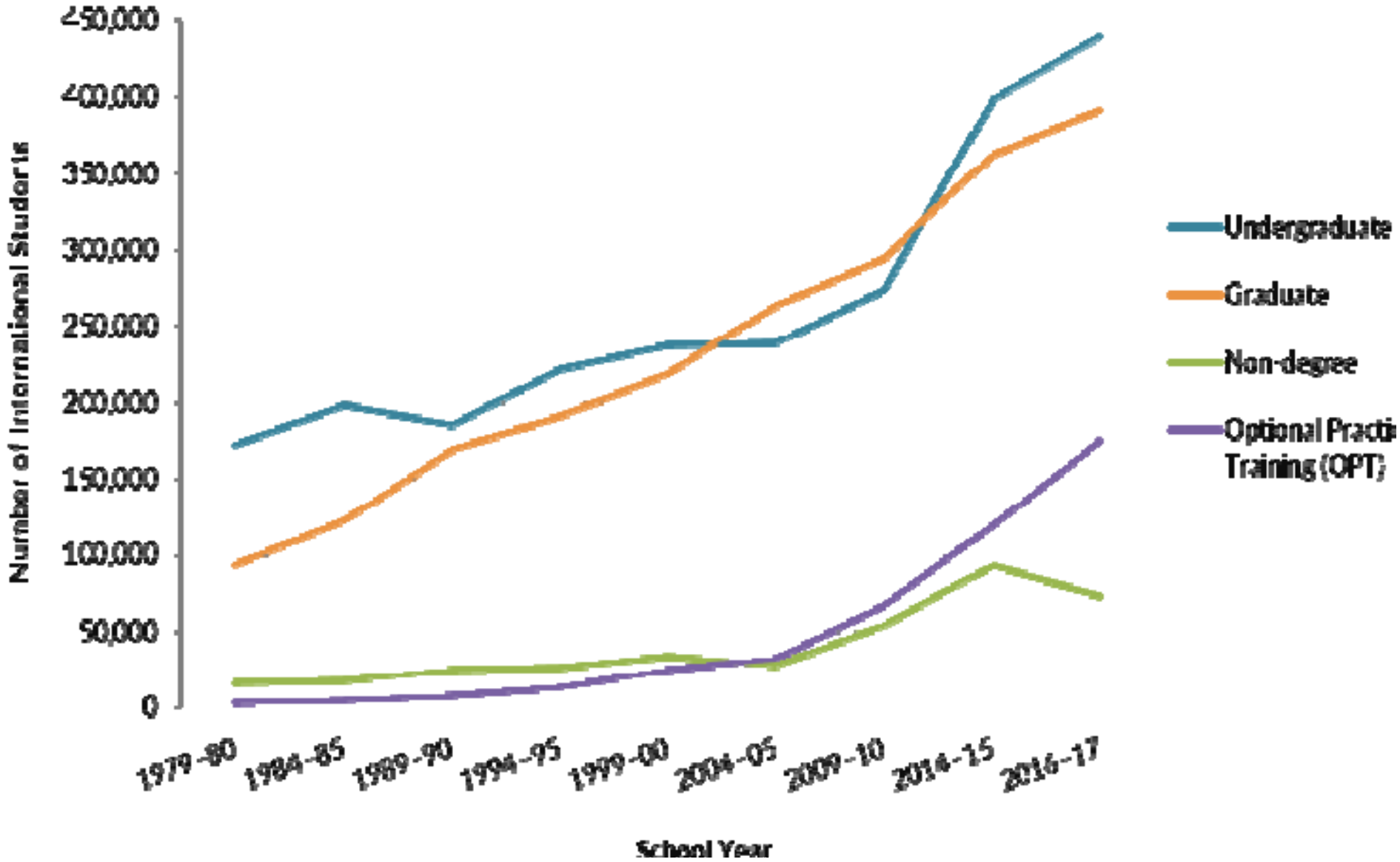
come from China and India.


Field of Study
2015/16
2016/17
Engineering
21%
Business and Management
19%
Math and Computer Science
15%
Other Fields of Study
8%
Social Sciences
8%
Physical and Life Sciences
7%
Fine and Applied Arts
6%
Intensive English
3%
Health Professions
3%
Communications and Journalism
2%
0K
50K
100K
150K
200K
Number of Students

Growth in selected fields of study of International students



Academic Level of International Students, SY 1979–80 to SY 2016–17



PRIMARY SOURCE OF FUNDING FOR INTERNATIONAL STUDENTS IN THE U.S.

personal & family

67%

U.S. college or university

17%

foreign government or university

7%

current employment

7%

$35.8 billion was contributed to the U.S. economy by international students in 2014/15. (Source: U.S. Department of Commerce)

Thank you!

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